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[Intervention Review]

Behavioural interventions for improving oral hygiene in adults with periodontal diseases

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ABSTRACT

Rationale

Periodontal diseases are common chronic inflammations in adults and can impact quality of life. Optimal oral hygiene behaviour is essential for establishing and maintaining periodontal health and ensuring treatment has the best chance of success. Therefore, it is important to establish effective ways to support adults with periodontal disease to maintain their oral hygiene.

Objectives

To determine the impact of behavioural interventions aimed at improving oral hygiene in adults with periodontal diseases (including gingivitis and periodontitis).

Search methods

We searched CENTRAL, MEDLINE, and three other databases up to 3 July 2024. We also searched two trials registers (26 February 2025) and reference lists of eligible studies and related systematic reviews.

Eligibility criteria

We included randomised controlled trials (RCTs) involving adults with periodontitis or gingivitis that evaluated interventions involving behaviour change techniques (BCTs) aimed at improving oral hygiene.

Outcomes

Our outcomes were bleeding, gingival inflammation, other clinical markers of periodontal disease (plaque, probing pocket depth (PPD), and clinical attachment loss (CAL)), and self-reported measures of oral health-related behaviours. We used any clinically relevant measure and the final follow-up time point reported by study authors.

Risk of bias

We used the Cochrane RoB 1 tool to assess risk of bias.

Synthesis methods

Given the nature of the data, we synthesised results using Synthesis Without Meta-analysis (SWiM) methods. We used GRADE to assess the certainty of evidence.

Included studies

We included 25 RCTs involving 1422 adults. Nineteen studies included only adults with periodontitis, and five only adults with gingivitis; one study included both disease types. Each study included different interventions, which we classified according to the Behaviour Change Technique (BCT) taxonomy (Michie and colleagues 2013) and described using BCT cluster labels. The evaluated interventions adopted a range of BCTs, including: goals and planning, feedback and monitoring, shaping knowledge, natural consequences, comparison of behaviour, associations, repetition and substitution, comparison of outcomes, reward and threat, regulation, antecedents, identity, scheduled consequences, and self-belief. Active controls included feedback and monitoring, shaping knowledge, comparison of behaviour, repetition and substitution, and antecedents.

Synthesis of results

We judged the certainty of evidence for all outcomes below to be very low due to risk of bias, inconsistency (between directions of effect), and imprecision (small sample sizes, or effect estimates including no benefit to either group).

Behavioural interventions compared with an active control for adults with periodontitis (17 studies, 892 participants)

Studies involved 18 to 337 participants with a follow-up of 3 weeks to 12 months.

For bleeding (measured as bleeding on probing (BOP)), two studies found that participants in the active treatment phase receiving a behavioural intervention had less BOP than those receiving control, but another study found little or no difference between groups. For participants undergoing supportive periodontal treatment (SPT), three studies found little or no difference between groups in BOP.

For gingival inflammation (GI), one study of participants in the active treatment phase found little or no difference between groups in GI, whilst another found less GI after behavioural interventions than control. For adults undergoing SPT, two studies found little or no difference between groups in GI.

For plaque, one study of participants in the active treatment phase found a benefit in favour of behavioural interventions; two found little or no difference between groups. Another study in the SPT phase found no benefit of the intervention. One study of participants in the active treatment phase found little or no difference between groups in PPD. At the time of our search, we found no published data for CAL or adverse effects.

For self-reported measures of behaviour, one study of participants in the active treatment phase found little or no difference between groups in toothbrushing frequency, but a benefit in favour of the intervention for toothbrushing duration and interdental cleaning frequency. Another study found more daily interdental cleaning after behavioural interventions. For adults undergoing SPT, one study found improved use of interdental cleaning devices (≥ 5 times a week) after behavioural interventions compared with control, but little or no difference between groups for toothbrushing (at least twice daily).

However, the evidence for all the above reported results is very uncertain.

Behavioural interventions compared with an active control for adults with gingivitis (6 studies, 958 participants)

Studies involved 22 to 538 participants with a follow-up of one to 12 months.

Three studies (one with two intervention groups) reported less gingival bleeding at end of follow-up in participants receiving a behavioural intervention than those receiving an active control. A cluster-RCT reported a greater reduction in gingival bleeding for participants with gingival bleeding $< 20\%$ at baseline in the behavioural interventions group than the active control group. However, one very small study reported very imprecise data for gingival bleeding, and there were similarly imprecise data for reduction in gingival bleeding for participants with gingival bleeding $\geq 20\%$ at baseline in the cluster-RCT.

One study found little or no difference between groups in GI; two studies reported imprecise data for plaque. One study found that gingival pocket depth was reduced after the intervention. No data were available for adverse effects. One study found little or no difference between groups in self-reported measures of at least twice-daily toothbrushing.

However, the evidence for all the above reported results is very uncertain.

Authors' conclusions

There is currently insufficient evidence for the effectiveness of behavioural interventions on clinical indicators of gingival and periodontal diseases. We are therefore unable to draw any conclusions about their effectiveness for improving oral hygiene in adults with periodontal diseases.

Future research should focus on interventions that have been developed based on plausible mechanisms of action to further our understanding of how interventions work to change behaviour. Future trials should seek to address the methodological limitations highlighted by this review. In addition, reporting of future trials should be comprehensive and transparent.

Funding

This Cochrane review was funded (in part) by University of Pennsylvania School of Dental Medicine.

Registration

Protocol available via DOI: 10.1002/14651858.CD012049

PLAIN LANGUAGE SUMMARY

What are the effects of behavioural approaches to oral hygiene advice for adults with gum disease?

Key messages

- For people with periodontitis (a severe form of gum disease), we are very uncertain whether behavioural approaches to oral hygiene advice have any important benefits when compared to usual advice alone.
- For people with gingivitis (a less severe form of disease), we are also very uncertain whether behavioural approaches to oral hygiene advice have any important benefits when compared to usual advice alone.
- Future studies are needed to better understand the role of behavioural approaches in supporting people with gum disease.

What is gum disease?

Symptoms of gum disease include bleeding and swollen gums and bad breath. Gingivitis is the earliest stage of the disease. If left untreated, gingivitis can lead to periodontitis, which is a more severe form of gum disease. Periodontitis can cause lasting damage to the soft tissues and bone around the teeth, and in some cases people may lose their teeth.

How is it treated?

People may need treatment for gum disease from their dentist or dental hygienist, sometimes including surgery or antibiotics. However, people can reduce the symptoms of gum disease by following good oral hygiene habits (toothbrushing at least twice a day and using floss or other devices to clean at the gum line between the teeth). Self-care is very important for people living with gum disease, and the success of any dental treatment depends on this. Although people with gum disease are given this oral hygiene advice from their dentist or dental hygienist, we know that some people still do not maintain good enough oral hygiene to manage their disease.

What did we want to find out?

We wanted to find out if behavioural approaches to oral hygiene advice are more effective than usual advice alone at improving oral hygiene habits in adults with gum disease. Behavioural approaches include behaviour change techniques that are designed to help people change their usual behaviour patterns.

In particular, we looked at bleeding (after gentle probing or pressure of the affected sites), inflammation (swollen gums), plaque (sticky film of bacteria that forms on the teeth), depth of pockets (spaces around the teeth caused by gum disease), and attachment of the tooth to the bone. We also looked at whether people were more likely to report improved oral hygiene habits, and whether there were any harms associated with behavioural approaches.

What did we do?

We searched for studies on adults with gum disease. We included studies that compared behavioural approaches to oral hygiene advice with usual advice alone. We compared and summarised the results according to whether the intervention was given to people who had periodontitis or gingivitis. For people with periodontitis, we reported the results separately for those who had never been treated for gum

disease (active treatment) or were receiving long-lasting care (supportive treatment). We rated our confidence in the evidence based on factors such as study methods and sizes.

What did we find?

We found 25 studies involving 1422 people. Nineteen studies involved only people with periodontitis; five involved only people with gingivitis; and one involved both disease types.

The included studies used different behavioural approaches to change people's oral hygiene habits. Types of approaches included using mobile phone apps (to provide oral hygiene information or to send reminders), intra-oral cameras or mirrors (to give people feedback on the progression of disease), providing more detailed and personalised advice, supplying oral hygiene treatment for home use, discussions using counselling or motivational interviewing techniques (to help change people's behaviours), or diary-keeping for later discussions in dental appointments.

Main results

For adults with periodontitis, it is unclear if behavioural approaches are better than usual advice alone at reducing bleeding, inflammation, plaque, or depth of pockets. It is also unclear if people receiving behavioural approaches are more likely to report improvement in their oral hygiene habits than those receiving usual advice alone. This finding was the same for people receiving active or supportive treatment for their disease.

For adults with gingivitis, it is unclear if behavioural approaches are better than usual advice alone at reducing the symptoms of gingivitis or improving people's oral hygiene habits.

At the time of our search, we found no studies that reported information about attachment of the tooth to the bone, or harms.

What are the limitations of the evidence?

We have very low confidence in the evidence in this review because:

- people may have been aware of the approach they were receiving, which could have affected the results;
- studies did not always report all the information that we needed;
- most studies included only very small numbers of people, or not enough people for us to be certain about the results;
- the studies were all designed differently, and there were sometimes differences between the results of the studies.

How up-to-date is this evidence?

This evidence is current to 3 July 2024.

SUMMARY OF FINDINGS

Summary of findings 1. Behavioural interventions compared with active control in adults with periodontitis

Behavioural interventions compared with active control in adults with periodontitis

Population: adults with diagnosed periodontitis in the active or supportive phase of treatment

Settings: dental clinics

Intervention: behavioural interventions, involving BCTs, that aim to change people's oral hygiene

Comparison: active control included the use of BCTs, including oral hygiene instruction, that would be expected in usual care but did not include the BCT intervention of interest

Outcome	Impact of the intervention (using data reported by study authors)	Number of participants (studies)	Certainty of the evidence (GRADE)	Comments
Bleeding Measured as percentage, using BOP Follow-up: end of study follow-up (ranging from 3 months to 12 months in included studies)	During active treatment: 1 study^a found less BOP in people exposed to a behavioural intervention than those exposed to an active control (MD -19.14%, 95% CI -30.90 to -7.38); another study^b also found less BOP in people exposed to a behavioural intervention than those exposed to an active control (MD -10.00%, 95% CI -15.09 to -4.91). Another study^c found little or no difference between intervention and control groups in BOP (MD 4.07%, 95% CI -5.18 to 13.32), and a further study^d also found little or no difference between intervention and control groups in BOP (MD 0.40%, 95% CI -7.05 to 7.85). During supportive treatment: 1 study^e found little or no difference between intervention and control groups in BOP (MD 1.60%, 95% CI -2.63 to 5.83); another study^f also found little or no difference between intervention and control groups in BOP (MD -3.00%, 95% CI -12.69 to 6.69); and a further study^g also found little or no difference between intervention and control groups in BOP (MD 2.40%, 95% CI -2.26 to 7.06).	337 participants (7 RCTs)	⊕⊕⊕⊕ Very low^h	
Gingival inflammation Measured using a 0-to-3 scale in which lower scores indicate lesser symptoms Follow-up: end of study follow-up (ranging from 1 month to 12 months in included studies)	During active treatment: 1 studyⁱ found little or no difference between intervention and control groups in gingival inflammation (MD -0.25, 95% CI -0.56 to 0.06). 1 study^b found fewer symptoms of gingival inflammation after exposure to a behavioural intervention (MD -0.29, 95% CI -0.35 to -0.23). During supportive treatment: 1 study^e found little or no difference between intervention and control groups in gingival inflammation (MD -0.16, 95% CI -0.33 to 0.01); another study^g also found little or no difference between intervention and control groups in gingival inflammation (MD -0.01, 95% CI -0.12 to 0.10).	245 participants (4 RCTs)	⊕⊕⊕⊕ Very low^j	

Clinical marker of periodontal disease: plaque Measured as a percentage Follow-up: end of study follow-up (ranging from 3 weeks to 12 months in included studies)	During active treatment: 1 study ^k found that participants exposed to a behavioural intervention had less plaque than those exposed to an active control (MD -20.00%, 95% CI -29.19 to -10.81); another study ^l found little or no difference between intervention and control groups in plaque (MD -4.84%, 95% CI -10.16 to 0.48); and a third study ^d also found little or no difference between intervention and control groups in plaque (MD 6.60%, 95% CI -1.88 to 15.08). During supportive treatment: 1 study ^e found no benefit in plaque measures for people exposed to a behavioural intervention (MD 8.50%, 95% CI 1.09 to 15.91).	191 participants (4 RCTs)	⊕⊕⊕⊕ Very low^h	We prioritised reporting data for plaque measured as a percentage. Some studies reported plaque using plaque index score; these data are described in the main review (without GRADE assessment).
Clinical marker of periodontal disease: PPD Measured in millimetres Follow-up: end of study follow-up (at 3 months in the included study)	During active treatment: 1 study ^a found little or no difference between intervention and control groups in PPD (MD -0.80 mm, 95% CI -2.12 to 0.52). During supporting treatment: no available evidence	18 participants (1 RCT)	⊕⊕⊕⊕ Very low^m	We prioritised reporting data for PPD measured in millimetres. Some studies reported the percent or number of pockets with varying depths; these data are described in the main review (without GRADE assessment).
Clinical marker of periodontal disease: CAL	No available evidence	-	-	
Self-reported measures of oral health-related behaviour Using different measures in each study Follow-up: end of study follow-up (ranging from 3 weeks to 12 months in included studies)	During active treatment: 1 study ^l found little or no difference between intervention and control groups in toothbrushing frequency (MD 0.18 times/day, 95% CI -0.22 to 0.58), but a benefit in favour of behavioural interventions when compared with an active control in toothbrushing duration (MD 1.78 minutes/day, 95% CI 0.64 to 2.92) and interdental cleaning frequency (MD 8.08 times/week, 95% CI 5.43 to 10.73). Another study ^b found that more people exposed to a behavioural intervention undertook daily interdental cleaning (RR 1.34, 95% CI 1.04 to 1.73). During supportive treatment: 1 study ^e found little or no difference between groups in toothbrushing frequency (MD -0.01, 95% CI -0.37 to 0.35) and interdental cleaning (MD 0.03, 95% CI -0.51 to 0.57), measured on a 5-point Likert scale with higher scores indicating more frequent cleaning. Another study ^g found that more people exposed to a behavioural intervention used interdental cleaning devices ≥ 5 times a week (RR 1.44, 95% CI 1.03 to 2.02) compared to those exposed to an active control, but found little or no difference between	244 participants (4 RCTs)	⊕⊕⊕⊕ Very lowⁱ	Another study reported categorical data for the performance of oral hygiene and for compliance to skills obtained during treatment. We did not calculate effect estimates for these data, which are described in the main review (without GRADE assessment).

	groups in $\geq$ twice-daily toothbrushing (RR 0.85, 95% CI 0.68 to 1.06).			
Adverse effects	No available evidence	-	-	No studies reported data specifically for adverse effects related to the intervention. 1 study reported that there were no adverse events during the study period.
Effects related to the interventions				

BCTs: behaviour change techniques; **BOP:** bleeding on probing; **CAL:** clinical attachment loss; **CI:** confidence interval; **MD:** mean difference; **OH:** oral hygiene; **PPD:** probing pocket depth; **RCT:** randomised controlled trial; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^aStudy by Hartono 2024. Intervention with BCTs including: feedback and monitoring, shaping knowledge, comparison of behaviour, associations, and repetition and substitution. Active control with BCTs including: shaping knowledge, comparison of behaviour, and repetition and substitution.

^bStudy by Jönsson 2009. Intervention with BCTs including: goals and planning, feedback and monitoring, shaping knowledge, comparison of behaviour, and repetition and substitution. Active control with BCTs including: shaping knowledge, comparison of behaviour, and repetition and substitution.

^cStudy by Järvinen 2020. Intervention with BCTs including: goals and planning, shaping knowledge, comparison of behaviour, repetition and substitution. Active control with BCTs including: shaping knowledge, comparison of behaviour, repetition and substitution.

^dStudy by Stenman 2012. Intervention with BCTs including: goals and planning, shaping knowledge, comparison of behaviour, associations, repetition and substitution, antecedents, identity, and self-belief. Active control with BCTs including: shaping knowledge, comparison of behaviour, and repetition and substitution.

^eStudy by Arnett 2023. Intervention with BCTs including: shaping knowledge, comparison of the behaviours, repetition and substitution, reward and threat, and scheduled consequences. Active control with BCTs including: shaping knowledge, comparison of behaviour, and repetition and substitution.

^fStudy by Brand 2013. Intervention with BCTs including: goals and planning, shaping knowledge, comparison of behaviour, repetition and substitution, comparison and outcomes, reward and threat, identity, and self-belief. Active control with BCTs including: shaping knowledge, comparison of behaviour, and repetition and substitution.

^gStudy by Jönsson 2006. Intervention with BCTs including: goals and planning, feedback and monitoring, shaping knowledge, natural consequences, comparison of behaviour, and repetition and substitution. Active control with BCTs including: shaping knowledge, comparison of behaviour, and repetition and substitution.

^hWe downgraded the certainty of the evidence by three levels: one level for risk of bias in each included study, one level for inconsistency (with differences in the direction of effect in each study), and one level for imprecision (with evidence mostly derived from studies with small sample sizes, and some CIs that included the possibility of benefit and no benefit of the intervention).

ⁱStudy by Alasmari 2022. Intervention with BCTs including: shaping knowledge, comparison of the behaviours, associations, and repetition and substitution. Active control with BCTs including: shaping knowledge.

^jWe downgraded the certainty of the evidence by two levels: one level for risk of bias in each included study, and two levels for imprecision (with evidence mostly derived from studies with small sample sizes, and some CIs that included the possibility of benefit and no benefit of the intervention).

^kStudy by Godard 2011. Intervention with BCTs including: goals and planning, feedback and monitoring, shaping knowledge, natural consequences, comparison of behaviour, repetition and substitution, antecedents, and self-belief. Active control with BCTs including: shaping knowledge, comparison of behaviour, repetition and substitution, and antecedents.

^lStudy by Kakudate 2009. Intervention with BCTs including: goals and planning, feedback and monitoring, shaping knowledge, comparison of the behaviour, repetition and substitution, and reward and threat. Active control with BCTs including: shaping knowledge, comparison of behaviour, and repetition and substitution.

^mWe downgraded the certainty of the evidence by three levels: one level for risk of bias, and two levels for imprecision (with evidence derived from a very small sample size, and CI that included the possibility of benefit and no benefit of the intervention).

Summary of findings 2. Behavioural interventions compared with active control in adults with gingivitis

Behavioural interventions compared with active control in adults with gingivitis

Population: adults with diagnosed gingivitis

Settings: dental clinics

Intervention: behavioural interventions, involving BCTs, that aim to change people's oral hygiene

Comparison: active control included oral hygiene instruction but involving fewer BCTs than the intervention. In the included studies, this typically more closely resembled usual care.

Outcome	Impact of the intervention (using data reported by study authors)	Number of participants (studies)	Certainty of the evidence (GRADE)	Comments
Gingival bleeding Measured as percentage, using bleeding on pressure at the gingival margin Follow-up: end of study follow-up (ranging from 1 to 6 months)	1 study ^a reported that people exposed to a behavioural intervention had less gingival bleeding than those in the control group (MD -29.00%, 95% CI -36.87 to -21.13). Another study ^b also found a benefit in favour of the intervention (MD -23.26%, 95% CI -32.46 to -14.06). 1 study ^c with 2 intervention groups found a benefit in favour of both types of intervention (MD -31.00%, 95% CI -45.19 to -16.81; MD -25.00%, 95% CI -36.65 to -13.35). 1 study ^d reported data that were very imprecise (MD -42.00%, 95% CI -86.67 to 2.67). 1 study ^e with separate data for baseline gingivitis reported the difference in change over time. For people with low levels of gingivitis, there was a larger improvement in gingival bleeding for people exposed to a behavioural intervention (MD -21.40, 95% CI -42.02 to -0.78). However, the findings for people with high levels of gingivitis were imprecise (MD -2.80, 95% CI -18.50 to 12.90).	701 participants (5 RCTs)	⊕⊕⊕⊕ Very low^f	
Gingival inflammation Measured using a 0-to-3 scale in which lower scores indicate lesser symptoms Follow-up: end of study follow-up (at 1 month in the included study)	1 study ^d found little or no difference between intervention and control groups in gingival inflammation (MD -0.70, 95% CI -1.66 to 0.26).	22 participants (1 RCT)	⊕⊕⊕⊕ Very low^f	

Clinical marker of periodontal disease: plaque Measured as a percentage Follow-up: end of study follow-up (ranging from 1 month to 3 months)	1 study ^d reported data for plaque that were very imprecise, with no apparent benefit for either the intervention or control group (MD -13.00%, 95% CI -28.47 to 2.47). 1 study ^c with 2 intervention groups also reported imprecise findings for plaque for both intervention types (MD -9.00%, 95% CI -21.41 to 3.41; MD -3.00%, 95% CI -11.49 to 5.49).	58 participants (2 RCTs)	⊕⊕⊕⊕ Very low^f	We prioritised reporting data for plaque measured as a percentage. One study reported plaque using an index score; these data are described in the main review.
Clinical marker of periodontal disease: depth of gingival pockets Measured in millimetres Follow-up: end of study follow-up (at 3 months in the included study)	1 study ^b found a reduction of gingival pocket depth in favour of the intervention (MD -1.33 mm, 95% CI -1.73 to -0.93).	22 participants (1 RCT)	⊕⊕⊕⊕ Very low^f	
Self-reported measures of oral health-related behaviours Using different measures as reported by study authors Follow-up: end of study follow-up (at 1 month in the included study)	1 study ^d found little or no difference between intervention and control groups in the number of participants brushing their teeth at least twice a day (RR 1.10, 95% CI 0.86 to 1.40).	22 participants (1 RCT)	⊕⊕⊕⊕ Very low^f	
Adverse effects Related to the intervention	No available evidence	-	-	No studies reported data specifically for adverse effects related to the intervention. 4 studies reported that there were no adverse events during the study period.

BCTs: behaviour change techniques; **CI:** confidence interval; **MD:** mean difference; **OH:** oral hygiene; **RCT:** randomised controlled trial; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^aStudy by Daly 2023. Intervention with BCTs including: feedback and monitoring, shaping knowledge, comparison of behaviour, associations, repetition and substitution, and antecedents. Active control with BCTs including: shaping knowledge, comparison of behaviour, and repetition and substitution.

^bStudy by Hartono 2024. Intervention with BCTs including: feedback and monitoring, shaping knowledge, comparison of behaviour, associations, and repetition and substitution. Active control with BCTs including: shaping knowledge, comparison of behaviour, and repetition and substitution.

^cStudy by Walsh 1985. Interventions (in both groups) with BCTs including: feedback and monitoring, shaping knowledge, natural consequences, comparison of behaviour, repetition and substitution, antecedents. Active control with BCTs including: feedback and monitoring, shaping knowledge, comparison of behaviour, repetition and substitution, antecedents.

^dStudy by Holloway 2022. Intervention with BCTs including: feedback and monitoring, shaping knowledge, natural consequences, comparison of behaviour, repetition and substitution, antecedents. Active control with BCTs including: shaping knowledge, comparison of behaviour, and repetition and substitution.

^eStudy by Holloway 2021. Data reported separately for people with low gingivitis (gingival bleeding < 20% at baseline) and high gingivitis (gingival bleeding ≥ 20% at baseline). Intervention with BCTs including: goals and planning, shaping knowledge, natural consequences, comparison of behaviour, repetition and substitution, comparison of outcomes, and antecedents. Active control with BCTs including: shaping knowledge, comparison of behaviour, and repetition.

^fWe downgraded the certainty of the evidence by three levels: one level for risk of bias in the included study, and two levels for imprecision (with evidence derived from studies with small sample sizes, and the CIs in some effect estimates including the possibility of benefit or no benefit for either group).

BACKGROUND

Description of the condition

Periodontal diseases affect the tissues that support teeth. The most common form is gingivitis. This is caused by the accumulation of dental plaque along the gum line causing inflammation, which can lead to swelling and bleeding [1]. Periodontitis is an inflammatory disease initiated and perpetuated by the subgingival microbial biofilm, leading to persisting chronic inflammation within the supporting tissues of the teeth; this can lead to progressive attachment loss and bone destruction. Globally, severe periodontitis affects about 19% of people over the age of 15 years, making it a considerable public health concern [2]. Periodontitis can impact negatively on oral health, speech, nutrition, self-confidence, and quality of life [3, 4, 5]. Achieving optimal oral hygiene is important for preventing periodontitis. Furthermore, the success of periodontal treatment depends on good long-term self-care. Periodontal treatment involves an active phase in which existing disease is treated and a supportive phase in which long-term health is maintained. It is possible that people may need different interventions to help them with their oral hygiene depending on which stage of treatment they are in. It is crucial to identify the most effective interventions for supporting individuals with periodontal disease in maintaining optimal long-term oral hygiene.

Description of the intervention and how it might work

Oral hygiene includes regular toothbrushing, interdental cleaning (such as flossing) or other actions to control levels of plaque. Interventions to support adults to achieve this should be focused on changing and maintaining these behaviours. This may take the form of instruction and demonstration, motivational interviewing or use of technology to support routines or provide feedback on behaviour amongst others.

Applying psychological and behavioural science (such as behaviour change techniques (BCTs)) to the design of such interventions will increase their effectiveness [6]. A BCT is defined as "an observable, replicable, and irreducible component of an intervention designed to alter or redirect causal processes that regulate behavior" [7]. BCTs are linked to theoretical constructs, such as stages of change or self-efficacy, and are considered the 'active components' of an intervention [8]. Michie and colleagues have created a taxonomy of 93 BCTs ("BCT taxonomy v1"). Examples include goal-setting, positive reinforcement, and social support [8]. The use of BCTs in oral hygiene instruction is likely to improve adherence [6].

Why it is important to do this review

Gingivitis can be prevented and reversed by optimal oral hygiene behaviour [5, 9], and the long-term success of treatment for periodontitis rests heavily on the performance of regular oral hygiene conduct. Supporting people with periodontal disease to maintain their own oral hygiene is vital, yet it remains unclear which intervention(s) are most effective. It is acknowledged that clinicians as well as the public consider this to be an important question [10, 11, 12, 13]. We have therefore conducted this Cochrane review of behavioural interventions to improve oral hygiene in adults with periodontal diseases.

OBJECTIVES

To determine the impact of behavioural interventions aimed at improving oral hygiene in adults with periodontal diseases (including gingivitis and periodontitis).

METHODS

We followed the Methodological Expectations of Cochrane Intervention Reviews (MECIR) when conducting the review [14], and PRISMA 2020 for the reporting [15]. Because we did not pool any data in the review, we also followed the nine items from the Synthesis Without Meta-analysis (SWiM) reporting criteria to complement PRISMA guidelines [16].

We previously published the Methods for this review in the Cochrane Library [17]. Whilst undertaking the review, we made minor changes in order to improve the clarity of reporting. In particular, we reported the results in distinct comparison groups according to disease type that we had not specified in advance. We also used more specific inclusion criteria, limiting studies only to those in which it was clear that all participants had periodontal disease and to interventions that included behaviour change techniques. We also changed the review title as this better reflected the current language for these intervention types. For a detailed description of changes from the protocol, see [Supplementary material 8](#).

Criteria for considering studies for this review

Types of studies

We included randomised controlled trials (RCTs) of parallel design. We also planned to include cluster-RCTs, where the unit of randomisation is at the level of a group. We excluded cross-over trials, which are inappropriate for this review because of the potential for carry-over or learnt effects, and we excluded quasi-randomised trials (in which allocation is not truly randomisation, such as date or birth or hospital record number).

Types of participants

We included adults (aged 18 years and over) diagnosed with periodontal disease. We included both diagnoses of gingivitis and periodontitis (any severity), with diagnosis by a clinician. For people with periodontitis, we included those who were in the active phase or supportive phase of treatment. We considered all types of participants, including those at higher risk of periodontal diseases (including gingivitis and periodontitis) such as pregnant women, smokers, and people with diabetes. In addition to people undergoing treatment for periodontal disease, we also planned to include the clinician as study participant, whereby the intervention attempts to change the behaviour of members of the dental team in order to improve periodontal outcomes for their dental patients.

We excluded studies in which not all participants had periodontal disease. Where this was not explicitly stated in the study criteria, we based this decision on the baseline data in the study report.

Types of interventions

We focused this review on interventions that aimed to change participants' oral health behaviours. We included any intervention with this aim that involved BCTs.

We reported review findings separately according to whether people had periodontitis or gingivitis and according to the type of comparison. We included the following comparisons in the review.

- Behavioural interventions compared with an active control for adults with periodontitis
- Behavioural interventions compared with no treatment for adults with periodontitis
- Behavioural interventions compared with behavioural interventions (using a different mode of delivery) for adults with periodontitis
- Behavioural interventions compared with an active control for adults with gingivitis
- Behavioural interventions compared with an alternative behavioural intervention for adults with gingivitis

For adults with periodontitis, we further stratified the data according to whether people were in the active or supportive phase of treatment. Had data been available, we would have also included a comparison with an alternative type of behavioural intervention for adults with periodontitis. We would have also included a comparison with no treatment and behavioural interventions (using a different mode of delivery) for adults with gingivitis.

Outcome measures

We excluded studies that had only a very short follow-up of less than one week because we considered this interval to be too short to determine whether participants had changed their oral hygiene behaviour.

In order to capture any benefit of an intervention over time, we collected outcome data at the final study follow-up.

Critical outcomes

- Bleeding. We distinguished between how this outcome is measured according to disease severity. Therefore, we reported percentage measures of bleeding on probing (BOP) for people with periodontitis, or measures of bleeding on pressure at the gingival margin for people with gingivitis as gingival bleeding. We also included scale measures of bleeding severity (such as a bleeding index scale).
- Gingival inflammation, measured on a scale (such as a gingival index scale).
- Any other clinical markers of periodontal diseases, for example plaque score, probing pocket depth (PPD) (or depth of gingival pockets for people with gingivitis), clinical attachment loss (CAL), recession, basic periodontal examination scores. For plaque, we included measures with or without disclosing solutions, reported as percentage or scale data.
- Self-reported measures of oral health-related behaviours.
- Adverse effects.

Gingival inflammation and gingival bleeding are both signs of gingivitis. However, inflammatory signs such as oedema (leading to gingival pocketing, or redness due to the changes in the subepithelial vessels) will be found earlier marked by evidence of 'bleeding by pressure'. Bleeding in gingivitis is a sign of increased inflammation in the gingival tissues. Although inflammatory signs without bleeding can also be found for periodontitis, this bleeding would occur because of probing of the pockets. Pocketing is a

result of attachment loss, and BOP is the result of the penetration of the blunt periodontal probe into the subepithelial granulation tissue. It can be difficult to distinguish between interactions of different dimensions of inflammation and bleeding, as these are highly associated to each other. In this review, we have reported these outcomes separately.

For studies that reported bleeding or plaque using different measures (an index scale or a percentage), we prioritised data reported as percentages. For PPD, we prioritised mean PPD measured in millimetres, rather than the per cent or number of varying depths. Although we reported both types of data in the review, we used the prioritised data when summarising these data in summary of findings tables and the review abstract.

Important outcomes

- Self-reported beliefs about and attitudes to oral health-related behaviour.

Search methods for identification of studies

Electronic searches

We searched the following databases for eligible trials:

- Cochrane Central Register of Controlled Trials (CENTRAL; 2024, Issue 7) in the Cochrane Library (searched 3 July 2024);
- MEDLINE Ovid (1946 to 3 July 2024);
- Embase Ovid (28 August 2015 to 3 July 2024);
- PsycINFO Ovid (1806 to 3 July 2024);
- Web of Science Conference Proceedings (1990 to 3 July 2024).

A former Information Specialist for the Cochrane Oral Health Group also searched Cochrane Oral Health's Trials Register (2 September 2022); this register is no longer kept up-to-date. As CENTRAL is updated monthly with newly published RCTs indexed in Embase and other databases, we restricted our search of Embase to recent years (2015 onwards). We did not place any other restrictions on the language or date of publication when searching the electronic databases. Subject strategies were modelled on the search strategy designed for MEDLINE Ovid. Where appropriate, these were combined with subject strategy adaptations of the highly sensitive search strategies designed by Cochrane for identifying RCTs (as described in the *Cochrane Handbook for Systematic Reviews of Interventions* [18]).

We also searched the following trial registries for ongoing studies:

- US National Institutes of Health Ongoing Trials Register ClinicalTrials.gov (clinicaltrials.gov; searched 26 February 2025);
- World Health Organization International Clinical Trials Registry Platform (apps.who.int/trialsearch; searched 26 February 2025).

For all search strategies, see [Supplementary material 1](#).

Searching other resources

When necessary, we contacted study authors of key papers and abstracts to request further information about their trials.

We searched the reference lists of included studies and relevant systematic reviews for further studies, and considered any additional eligible studies that were published in indexed journals.

We did not perform a separate search for adverse effects of interventions used; we considered adverse effects described in the included studies only.

To identify if any eligible studies had been retracted, we ran searches in Retraction Watch using reference information in eligible studies.

Data collection and analysis

Selection of studies

At least two review authors (PA, MB, ME, P-MJ-S, SL, LO'M, PP) independently and in duplicate screened all titles and abstracts identified by the search for potential eligibility, resolving any discrepancies through discussion with the review team. We obtained the full-text articles for those studies deemed potentially eligible, and at least two review authors (P-MJ-S, SL, LO'M, PP, PR) independently and in duplicate assessed the full texts to determine eligibility; again, we resolved disagreements through discussion with the review team. We used Covidence to manage the screening process [19].

Data extraction and management

At least two review authors (P-MJ-S, LO'M, SL, PR) independently and in duplicate conducted data extraction using a data extraction form that had been piloted, reaching consensus through discussion with the review author team. We used Covidence to support the data extraction process [19], and include a template data extraction form in [Supplementary material 9](#).

Two review authors (ME, GK) independently and in duplicate coded the components of the interventions and controls in each study using the BCT taxonomy v1 [7]; we report the full taxonomy in [Supplementary material 10](#). Any disagreements were resolved by discussion with a third review author (PA). In addition, we used the COMPASS checklist to describe interventions; this offers a systematic approach including specific domains (context, content, mechanism, outcome, mode of delivery) [20].

We translated study reports in languages other than English through translators connected to the Cochrane Oral Health Group and members of our own author team. We attempted to retrieve missing data from the original study authors.

Risk of bias assessment in included studies

Two review authors (LO'M, SL, PR) independently and in duplicate assessed the risk of bias for studies that contributed outcome data to the review using the Cochrane RoB 1 tool [21]. A third review author was available to resolve any disagreements in risk of bias assessments.

We assessed each included study for risk of bias in seven key domains.

- Sequence generation (selection bias)
- Allocation concealment (selection bias)
- Blinding of participants and personnel (performance bias)
- Blinding of outcome assessment (detection bias)
- Incomplete outcome data (attrition bias)
- Selective reporting (reporting bias)
- Other bias

We judged detection bias separately according to whether the outcomes were clinically assessed or self-reported by the study participants. For each risk of bias domain, we judged whether risk of bias in the study was low, high, or unclear.

Measures of treatment effect

For dichotomous outcomes (e.g. number of participants who carried out an oral health-related behaviour), we expressed the estimate of effect as a risk ratio (RR) together with 95% confidence interval (CI). For continuous outcomes, we employed means and standard deviations in the study reports to summarise the data for each group using mean differences (MDs) and 95% CIs.

Had we combined data from more than one study in the review, we would have calculated standardised mean differences for outcomes that had used different tools to measure the same outcome (e.g. different gingivitis or plaque scales).

We did not calculate effect estimates for categorical data, in which study authors reported two or more possible categories (or responses) to self-reported measures of oral health-related behaviours or beliefs and attitudes to oral health-related behaviour; instead, we presented the data in tables.

Unit of analysis issues

In the event of including multi-arm studies, we reported data separately for each eligible comparison; we did not pool data across study arms.

Dealing with missing data

Where information was missing from published reports, we attempted to retrieve the information from the original study authors.

Where data were missing and could not be obtained, we used methods for estimating missing standard deviations as described in Chapter 6 of the *Cochrane Handbook for Systematic Reviews of Interventions* [22].

Reporting bias assessment

Had we combined data, we would have assessed publication bias in any meta-analysis with sufficient studies according to the recommendations on testing for funnel plot asymmetry [23, 24].

Synthesis methods

We did not conduct meta-analysis or pool the studies together because studies were not comparable to one another in terms of intervention design. Therefore, we used SWiM methods according to Campbell and colleagues [16]. We summarised data separately for each comparison according to the type of comparison, whether people had periodontitis or gingivitis, and the phase of their treatment (see [Types of interventions](#) and [Investigation of heterogeneity and subgroup analysis](#)). For each outcome in each study, we calculated an effect estimate which we presented in forest plots and described in the main text with direction of effect as appropriate. If we could not present data in forest plots (such as categorical data), we presented these results in Additional tables.

Investigation of heterogeneity and subgroup analysis

We did not conduct formal subgroup analysis because we did not undertake meta-analysis. However, we collected the following information from studies and considered this information when describing the eligible studies in this review.

- Participant age: 18 to 30 years versus ≥ 30 years.
- Number of BCTs: we based this on the median number of BCTs in the intervention groups.
- Disease type: periodontitis or gingivitis.
- Phase of treatment for people with periodontitis (active or supportive).

We considered the latter two criteria (disease type and phase of treatment) to be distinct population groups and therefore reported data separately according to these criteria as appropriate throughout the review (see also [Types of interventions](#)),

We had planned to consider the combinations of BCTs, but because studies were so widely different, it was not possible to determine a suitable strategy for describing this. We also planned to consider the exposure to the intervention (particularly in terms of intensity, such as how many sessions or the duration of sessions). Although we described this information in supplementary materials, it was similarly not possible to determine a suitable strategy for describing this in quantifiable terms.

Equity-related assessment

We did not plan to assess equity-related issues in this review. However, as a change from protocol, we collected equity-related information from study reports based on the PROGRESS-Plus criteria, relating to place of residence, race/ethnicity/culture/language, occupation, gender/sex, religion, education, socioeconomic status, social capital, personal characteristics associated with disability, features of relationships, and time-dependent relationships [25]. We reported this information in [Supplementary material 13](#) and the overview tables ([Table 1](#); [Table 2](#)).

Sensitivity analysis

We did not conduct sensitivity analysis because we did not undertake meta-analysis.

Certainty of the evidence assessment

We constructed summary of findings tables including the critical outcomes measured at the longest follow-up for the comparisons that provided the most valuable information to users in contemporary practice (i.e. by comparing an intervention with an active control), as follows.

- Behavioural interventions compared with active control for adults with periodontitis
- Behavioural interventions compared with active control for adults with gingivitis

We used the GRADE approach for this process, which considers five domains: within-study risk of bias, indirectness, inconsistency, imprecision, and publication bias. We used the methods and recommendations described in Chapter 14 of the *Cochrane Handbook for Systematic Reviews of Interventions* to guide the application of GRADE methodology [26].

For other clinical markers of periodontal disease, we only reported data in the summary of findings tables for plaque, PPD, and clinical attachment loss for people with periodontitis. For people with gingivitis, we reported plaque and depth of gingival pockets. Some outcomes were reported using different measures. For both bleeding and plaque, we prioritised data reported as percentages rather than data reported on scales, and we indicated in the summary of findings table that other data are available in the main report. For self-reported measures of oral health-related behaviours, we reported all measures described by study authors. As the review did not include meta-analysis, we described data in the summary of findings tables using the SWiM approach [16].

Consumer involvement

We did not involve consumers in the development or conduct of this review.

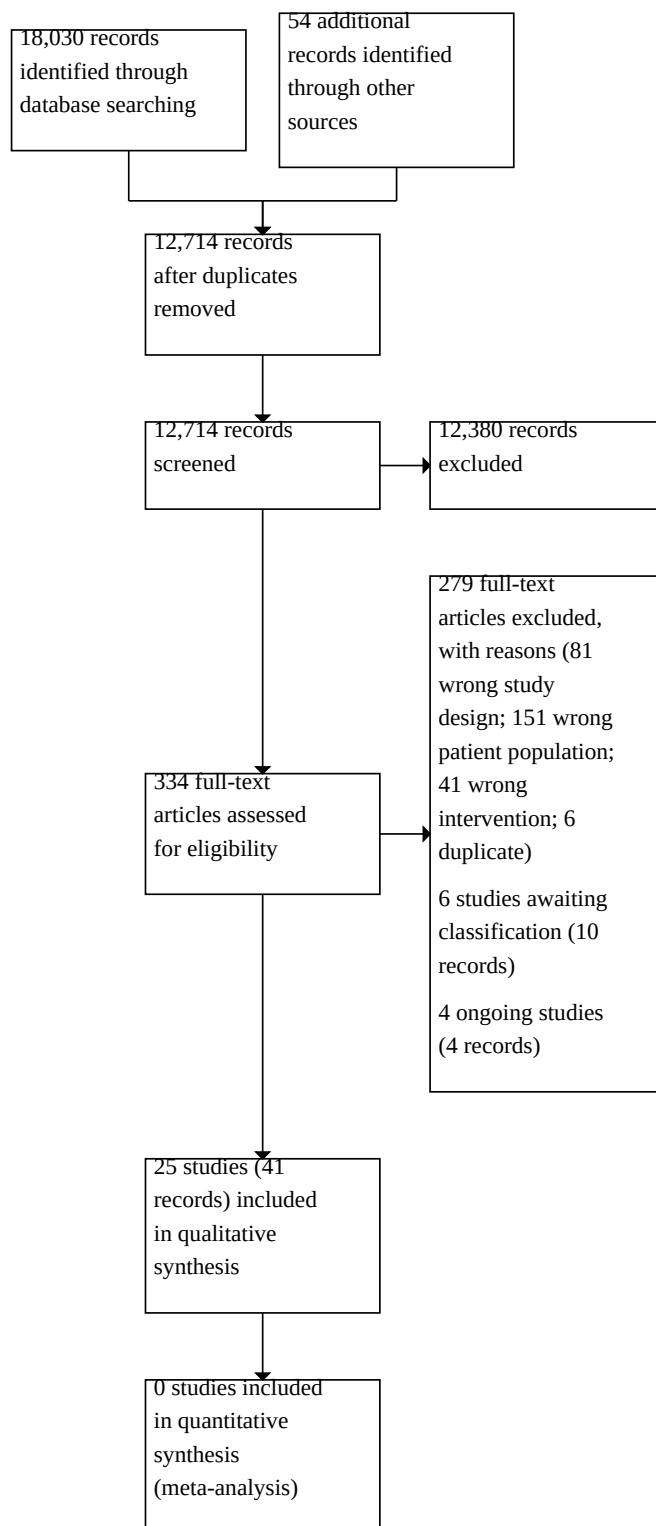
RESULTS

Description of studies

Results of the search

The database searches identified 18,030 records. [Figure 1](#) shows the study screening process. We included 25 studies after full-text review.

Figure 1. Study flow diagram.



Included studies

The 25 RCTs in this review involved 1422 randomised participants. Details of study characteristics, including setting, funding sources, and conflicts of interest reported by study authors, are provided in [Supplementary material 2](#). We contacted the study authors of one study for additional data as required (Arnett 2023 [27, 28, 29, 30, 31, 32]).

Nineteen studies involved only people who had periodontitis, and five studies involved only people with gingivitis. One study involved people with both periodontitis and gingivitis, and we were able to include participants from this study in two comparison groups because data were reported separately according to disease (Hartono 2024 [33, 34]).

Studies were conducted in Europe, the USA, Asia, and South America. They were mostly single-centre studies in university dental clinics.

In this review, we reported data separately according to the type of comparison and the disease status of participants, as follows. We found wide variation between intervention designs, with no studies using the same behavioural interventions. We present more detailed descriptions of the interventions and the BCTs involved in these interventions in [Supplementary material 11](#).

Behavioural interventions compared with active control for adults with periodontitis

This comparison included 17 studies involving 892 participants. Most studies included participants in the active phase of periodontal treatment. Some interventions involved technology (in particular, the use of mobile phone apps). Other interventions included additional oral hygiene instruction (with or without additional information resources), supply of oral hygiene resources for home use, and behavioural approaches to oral hygiene discussions such as motivational interviewing, diary or record keeping. The active control in studies typically involved oral hygiene instructions that were similar to usual care practice (sometimes called a 'tell-show-do' approach). There was a median of seven additional BCTs in the intervention groups than in the control groups (ranging from two to 18 more additional BCTs).

[Table 1](#) provides a brief overview of these studies, including outcomes and time of study follow-up.

Behavioural interventions compared with no treatment for adults with periodontitis

This comparison included one study involving 116 participants (Lhakang 2016 [35]). The intervention included strategies targeting self-efficacy and self-regulatory skills. This study reported important outcomes but no critical outcomes in this review, and these were reported at three weeks.

Behavioural interventions compared with behavioural interventions (using a different mode of delivery) for adults with periodontitis

This comparison included two studies involving 128 participants in the active phase of treatment for periodontitis. In Cleeren 2014 [36], information was conveyed to participants either using 3D animations or using real-time sketches that were recorded and used in a video to convey the same information. This study

reported adverse effects related to the intervention but no other relevant review outcomes. In Williams 2018 [37], one group used an audiovisual PowerPoint presentation on a computer to convey information, whilst an instructor delivered the same information (using the same script and visual aids) in the other group. Williams 2018 reported data for bleeding and plaque at a one-month follow-up.

Behavioural interventions compared with active control for adults with gingivitis

This comparison included six studies involving 958 participants with gingivitis; most participants were involved in one large study (Holloway 2021 [38]). In this large study, separate groups were randomised according to whether they had low levels of gingivitis (gingival bleeding < 20% of total sites at baseline) or high levels of gingivitis (gingival bleeding ≥ 20% of total sites at baseline). One study included two intervention groups, and we were able to include data for both groups separately compared with the control (Walsh 1985 [39]). Some interventions involved technology (intra-oral cameras for feedback and monitoring). Other interventions included additional oral hygiene instruction (with or without additional information resources or oral hygiene resources) or behavioural approaches to discussions about oral hygiene. The active control was similar to usual care practice. There was a median of five additional BCTs in the intervention group than in the control group (ranging from four to eight additional BCTs).

[Table 2](#) provides a brief overview of these studies, including outcomes and time of study follow-up.

Behavioural interventions compared with an alternative behavioural intervention for adults with gingivitis

This comparison included one study involving 24 participants (Walsh 1985). Participants in both intervention groups were instructed in the use of a toothpick for interproximal cleaning, with one group being advised to note the presence of bleeding to indicate the need for additional oral hygiene. This study only reported data for bleeding on probing and plaque, both with a three-month follow-up.

Excluded studies

We excluded 279 articles at full-text review. See [Supplementary material 3](#) for details of 13 key excluded studies (studies that a knowledgeable reader might reasonably expect to find in the review). We excluded two studies because they used an ineligible study design, one study because we could not determine whether all participants had periodontal disease, nine studies owing to intervention design (in four studies, BCTs were not present or this could not be determined, or there was no difference in the BCTs applied across intervention and control groups; in four studies, a powered toothbrush was provided to participants in the intervention group that we believed could have impacted the results; and in one study, the intervention was not directed at the study participants), and one study because it reported no follow-up time between delivery of the intervention and outcome measurements.

Studies awaiting classification

Six studies are awaiting classification; see [Supplementary material 4](#). Three studies were published only as abstracts with insufficient information to permit a full assessment of eligibility (Allawati

2022 [40]; Lin 2012 [41]; Zenobia 2020 [42]). There was insufficient information in two published studies to determine eligibility, and we await further information from the authors of these studies (Araújo 2016 [43, 44, 45]; Donos 2020 [46, 47]). One study, identified as ongoing in database searches, was published after completion of this review, and we will include it in future updates (Tytgat 2025 [48, 49]).

Ongoing studies

We identified four ongoing studies that are likely to be eligible for this review; see [Supplementary material 5](#). These studies all

compared behavioural interventions with an active control in adults with periodontitis, with one study also including adults with gingivitis. The total estimated recruitment for these studies is 468 participants.

Risk of bias in included studies

We have included risk of bias assessments and support for judgements for each study in [Supplementary material 2](#), with a summary figure in [Figure 2](#). One study reported no data for our review outcomes, and we did not conduct a risk of bias assessment for this study (Zhang 2018 [50]).

Figure 2. Risk of bias summary. Assessment was conducted only where outcome data were available; blank spaces indicate risk of bias assessment was not completed.

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias): Clinically-assessed outcomes	Blinding of outcome assessment (detection bias): Self-reported outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
Alasmari 2022	?	?	-	-		?	?	+
Arnett 2023	+	?	+	+	-	+	?	+
Baab 1986	+	?	+	+	-	+	?	+
Brand 2013	+	?	-	+	-	+	?	+
Cleeren 2014	+	+	+	-		+	?	+
Daly 2023	+	+	+	+		+	?	+
de Souza 2002	?	?	-	?	-	+	?	+
Fiebranz 1989	?	?	+		-	+	?	+
Godard 2011	+	+	-	+		+	?	+
Gunpinar 2022	+	?	-	+	-	+	?	+
Hartono 2024	+	+	-	+		+	?	+
Holloway 2021	+	+	-	?	-	?	+	+
Holloway 2022	+	-	-	+		+	+	+
Järvinen 2020	?	?	-	?		-	?	+
Jönsson 2006	+	?	+	+	-	+	?	+
Jönsson 2009	+	+	+	+	-	+	?	+

Figure 2. (Continued)

Jönsson 2006	+	?	+	+	-	+	?	+
Jönsson 2009	+	+	+	+	-	+	?	+
Kakudate 2009	+	?	+	+	-	+	-	+
Lhakhang 2016	?	?	-		-	+	?	+
Lin 2019	-	-	-		-	-	?	+
Little 1997	?	?	-	+	-	+	?	+
Philippot 2005	?	?	-	?		?	?	+
Stenman 2012	+	+	-	+		+	?	+
Walsh 1985	?	?	+	+	-	?	?	+
Williams 2018	?	?	+	+		+	?	+
Zhang 2018								

Whilst most studies reported an adequate method for randomisation, information about allocation concealment was often insufficiently reported. We judged only seven studies to be at low risk of selection bias for both random sequence generation and allocation concealment (Cleeren 2014; Daly 2023 [51]; Godard 2011 [52]; Hartono 2024; Holloway 2021; Jönsson 2006 [53]; Stenman 2012 [54, 55]). We judged Lin 2019 [56] to be at high risk of selection bias because of concerns about an uneven balance of baseline characteristics in this large trial.

Because of the nature of the interventions, it was not possible to blind participants to their allocation; in studies with an active control considered to be over and above usual care, we considered the risk of performance bias to be low (Arnett 2023; Baab 1986 [57]; Cleeren 2014; Daly 2023; Fiebranz 1989 [58]; Jönsson 2006; Jönsson 2009 [59, 60, 61, 62, 63, 64, 65]; Kakudate 2009 [66]; Walsh 1985; Williams 2018); otherwise, we judged performance bias to be at high risk. The risk of detection bias was high across all studies for self-reported outcome measures; however, we found that most studies included adequate methods to ensure blinding for the assessment of clinical outcomes such that we judged them to be at low risk of detection bias. We judged only Alasmari 2022 [67] and Cleeren 2014 to be at high risk of detection bias for clinically assessed outcomes.

Whilst most studies were at low risk of attrition bias, four studies reported insufficient information for us to be confident that data were reported for all participants (Alasmari 2022; Holloway 2021; Philippot 2005 [68]; Walsh 1985). We judged the risk of attrition bias to be high in Järvinen 2020 [69, 70] and Lin 2019 owing to a large imbalance in losses between groups, and a large number of losses that were not explained in Lin 2019. However, all other studies reported sufficient information to indicate that there were no losses, or losses were explained, balanced between groups, and likely too few to impact the study findings.

Only three studies reported prospective clinical trials registrations (Arnett 2023; Holloway 2021; Holloway 2022 [71, 72]). The other studies either did not report clinical trials registrations or protocols, or registration was completed retrospectively, meaning we were

unable to effectively judge the risk of selective reporting bias. Only Kakudate 2009 was judged to be at high risk of selective reporting bias.

We identified no other sources of bias in the included studies.

Synthesis of results

Behavioural interventions compared with an active control for adults with periodontitis

Due to the presence of statistical heterogeneity when initially pooling the data, likely owing to the wide differences between each intervention design, we did not present results of meta-analyses, but instead reported a narrative summary of the effect estimates for each outcome.

Critical outcomes

For all critical outcomes, we downgraded the certainty of the evidence by one level for risk of bias in each included study. For the outcomes of bleeding and plaque, we downgraded the certainty of the evidence by one level for inconsistency due to differences in the direction of effect in each study, and by one level for imprecision due to evidence being mostly derived from studies with small sample sizes and some CIs that included the possibility of benefit and no benefit. For the outcomes of gingival inflammation, PPD, and self-reported outcomes, we downgraded the certainty of the evidence by two levels for imprecision because the evidence was mostly derived from studies with small or very small sample sizes, and some CIs included the possibility of benefit and no benefit. See [Summary of findings 1](#).

Two studies reported data for which we were unable to calculate effect estimates (e.g. because of missing SDs) (Fiebranz 1989; Little 1997 [73]); see [Supplementary material 12](#). We did not include these data when assessing the certainty of the evidence.

Bleeding

See Analysis 1.1.

For participants in the **active phase of treatment**, Hartono 2024 found that participants exposed to a behavioural intervention had less BOP than participants exposed to an active control (MD -19.14%, 95% CI -30.90 to -7.38; 18 participants). Jönsson 2009 also found a benefit in favour of the behavioural intervention (MD -10.00%, 95% CI -15.09 to -4.91; 108 participants). However, there was little or no difference between the intervention and control groups in BOP in Järvinen 2020 (MD 4.07%, 95% CI -5.18 to 13.32, favours control; 21 participants), and Stenman 2012 (MD 0.40%, 95% CI -7.05 to 7.85, favours control; 44 participants).

For participants in the **supportive phase of treatment**, there was no evidence of a difference between intervention and active control groups in BOP in any of the following three studies: Arnett 2023 (MD 1.60%, 95% CI -2.63 to 5.83, favours control; 58 participants); Brand 2013 [74] (MD -3.00%, 95% CI -12.69 to 6.69, favours intervention; 53 participants); Jönsson 2006 (MD 2.40%, 95% CI -2.26 to 7.06, favours control; 35 participants).

Gingival inflammation

Studies measured this outcome using the 0-to-3 scale of Löe and Silness, in which lower scores indicate lesser symptoms [75]; see Analysis 1.2.

For participants in the **active phase of treatment**, Alasmari 2022 found little or no difference between the intervention and control groups in gingival inflammation (MD -0.25, 95% CI -0.56 to 0.06, favours intervention; 39 participants). However, Jönsson 2009 found that participants had less gingival inflammation after exposure to a behavioural intervention (MD -0.29, 95% CI -0.35 to -0.23; 113 participants).

For participants in the **supportive phase of treatment**, two studies found little or no difference between intervention and control groups in gingival inflammation: Arnett 2023 (MD -0.16, 95% CI -0.33 to 0.01, favours intervention; 58 participants); Jönsson 2006 (MD -0.01, 95% CI -0.12 to 0.10, favours intervention; 35 participants).

Other clinical markers of periodontal disease

Plaque

This was reported using two measures, either as a percentage (Analysis 1.3) or a scale (Analysis 1.4).

In the **active phase of treatment**, when plaque was measured as a percentage, Godard 2011 found that participants exposed to a behavioural intervention had less plaque than those exposed to an active control (MD -20.00%, 95% CI -29.19 to -10.81, favours intervention; 51 participants). In Kakudate 2009, there was little or no difference between groups in plaque (MD -4.84%, 95% CI -10.16 to 0.48, favours intervention; 38 participants). There was also little or no difference between groups in Stenman 2012 (MD 6.60%, 95% CI -1.88 to 15.08, favours control; 44 participants).

Only one study reported plaque as a percentage during the **supportive phase of treatment** (Arnett 2023), finding that behavioural interventions provided no benefit to plaque (MD 8.50%, 95% CI 1.09 to 15.91, favours control; 58 participants).

Five studies measured plaque using a scale, either the 0-to-3 scale of Silness and Löe [76] (Jönsson 2006; Jönsson 2009; Philippot 2005) or the 0-to-5 plaque scale of Quigley-Hein Turesky

Modification Index [77] (Alasmari 2022; Brand 2013); in both scales, lower scores indicate less plaque.

In the **active phase of treatment**, when plaque was measured on a scale, Alasmari 2022, found little or no difference between groups in plaque (MD -0.21, 95% CI -0.48 to 0.06, favours intervention; 39 participants). However, Jönsson 2009 found that participants exposed to a behavioural intervention had less plaque than those exposed to an active control (MD -0.17, 95% CI -0.22 to -0.12; 113 participants). Philippot 2005 also found a benefit in favour of the intervention (MD -0.64, 95% CI -0.86 to -0.42; 30 participants).

In the **supportive phase of treatment**, when plaque was measured on a scale, Brand 2013 found little or no difference between groups (MD -0.20, 95% CI -0.58 to 0.18, favours intervention; 53 participants). Jönsson 2009 found a small benefit to participants exposed to a behavioural intervention (MD -0.08, 95% CI -0.15 to -0.01; 35 participants).

Probing pocket depth (PPD)

One study reported PPD (Hartono 2024). Participants in this small study were in the **active phase of treatment**, and the authors found little or no difference in PPD between groups (MD -0.80 mm, 95% CI -2.12 to 0.52, favours intervention; 18 participants; Analysis 1.5).

Other studies reported the per cent or number of pockets with varying depths, and all studies found little or no difference in effect between groups; see Analysis 1.6. Jönsson 2009 reported data, involving participants in the **active phase of treatment**, for the per cent of pockets with a PPD of 4 to 5 mm (MD -1.80%, 95% CI -5.36 to 1.76, favours intervention; 108 participants) and the per cent of pockets with PPD $\geq$ 6 mm (MD 0.10%, 95% CI -1.05 to 1.25, favours control; 108 participants). In the **supportive phase of treatment**, Brand 2013 reported data for the per cent of pockets with PPD of 4 to 6 mm (MD 4.2%, 95% CI -5.95 to 14.35, favour control; 53 participants) and for the per cent of pockets with PPD of $\geq$ 7 mm (MD 0.30%, 95% CI -2.39 to 2.99, favours control; 53 participants), and Jönsson 2006 reported the number of pockets with PPD of $>$ 4 mm (MD -0.20, 95% CI -2.23 to 1.83, favours intervention; 35 participants).

Self-reported measures of oral health-related behaviours

In participants in the **active phase of treatment**, Kakudate 2009 found little or no difference between intervention and control groups in toothbrushing frequency (MD 0.18 times/day, 95% CI -0.22 to 0.58, favours intervention; 38 participants), but a benefit in toothbrushing duration (MD 1.78 minutes/day, 95% CI 0.64 to 2.92; 38 participants) and interdental cleaning frequency (MD 8.08 times/week, 95% CI 5.43 to 10.73; 38 participants) for participants exposed to a behavioural intervention compared to those exposed to an active control; see Analysis 1.7. Jönsson 2009 found that more participants exposed to a behavioural intervention were likely to undertake daily interdental cleaning than those exposed to an active control (RR 1.34, 95% CI 1.04 to 1.73; 113 participants; Analysis 1.8).

In addition, Jönsson 2009 reported categorical data for performance of oral hygiene (much better or better) and compliance to skills obtained during treatment (daily, several times a week, some times a week). These data are reported in Table 3.

In participants in the **supportive phase of treatment**, Arnett 2023 found little or no difference between intervention and control groups in toothbrushing frequency (MD -0.01, 95% CI -0.37 to 0.35, favours control; 58 participants) and interdental cleaning frequency (MD 0.03, 95% CI -0.51 to 0.57, favours intervention; 58 participants); these outcomes were measured on a 5-point Likert scale, with higher scores indicating more frequent cleaning (Analysis 1.7). Jönsson 2006 found that more participants exposed to a behavioural intervention were likely to have improved use of interdental cleaning devices (at least five times per week) than those exposed to an active control (RR 1.44, 95% CI 1.03 to 2.02; 35 participants), but that there was little or no difference between intervention and control groups in the number of people who performed at least twice-daily toothbrushing (RR 0.85, 95% CI 0.68 to 1.06, favours control; 35 participants); see Analysis 1.8.

Adverse effects

No studies reported adverse effects related to the intervention. However, we noted that Hartono 2024 reported that: "All patients successfully completed the study in an uneventful manner and without any adverse events".

Important outcomes

Self-reported beliefs about and attitudes to oral health-related behaviour

For participants in the **active phase of treatment**, Kakudate 2009 found that self-efficacy for brushing teeth was improved for those in the behavioural intervention group compared with the active control group (MD 3.16, 95% CI 1.55 to 4.77; 38 participants), but de Souza 2002 [78] found little or no difference between groups in self-efficacy levels (MD 4.43, 95% CI -3.44 to 12.30, favours intervention; 60 participants). See Analysis 1.9.

For participants in the **supportive phase of treatment**, Arnett 2023 (with 58 participants) found little or no difference between groups in self-efficacy for: performing oral hygiene (MD 0.11, 95% CI -0.31 to 0.53, favours intervention); making a change to improve oral health (MD 0.02, 95% CI -0.41 to 0.45, favours intervention); and maintaining changes to oral hygiene (MD 0.15, 95% CI -0.31 to 0.61, favours intervention). In addition, Arnett 2023 found little or no difference between groups in perceived importance of good oral health (MD 0.06, 95% CI -0.20 to 0.32, favours intervention) and good oral hygiene (MD 0.02, 95% CI -0.26 to 0.30, favours intervention), as well as in levels of interest in oral self-care (MD 0.12, 95% CI -0.7 to 0.31, favours intervention). Brand 2013 found little or no difference between groups in scores of motivation (MD 3.20, 95% CI -0.09 to 6.49, favours intervention; 53 participants) and in controlled self-regulation (MD 0.60, 95% CI -3.55 to 4.75, favours intervention; 53 participants). However, there was a benefit in scores of participant's autonomous regulation after exposure to a behavioural intervention (MD 2.80, 95% CI 1.06 to 4.54; 53 participants). See Analysis 1.9.

In addition, Fiebranz 1989 and Jönsson 2009 reported categorical data in response to self-reported questionnaires, which are reported in Table 3.

Behavioural interventions compared with no treatment for adults with periodontitis

Only Lhakhang 2016 reported relevant outcome data; this included data at only a follow-up of three weeks. No outcome data

were available for bleeding, gingival inflammation, other clinical markers of periodontal disease, or adverse effects.

Critical outcomes

Self-reported measures of oral health-related behaviours

Lhakhang 2016 reported that flossing frequency (measured as number of times participants flossed per day) was slightly higher in participants exposed to a behavioural intervention (MD 0.87, 95% CI 0.52 to 1.22; 112 participants; Analysis 2.1).

Important outcomes

Self-reported beliefs about and attitudes to oral health-related behaviour

Lhakhang 2016 reported that the number of times per day that participants intended to floss was higher for those exposed to a behavioural intervention (MD 0.76, 95% CI 0.35 to 1.17; 112 participants). However, there was no evidence of a difference in flossing self-efficacy (MD 0.18, 95% CI -0.18 to 0.54, favours intervention; 112 participants). See Analysis 2.2.

Behavioural interventions compared with behavioural interventions (using a different mode of delivery) for adults with periodontitis

No outcome data were available for gingival inflammation, clinical markers of periodontal disease other than plaque, self-reported measures of oral health-related behaviours, or self-reported beliefs and attitudes to oral health-related behaviour.

Critical outcomes

Bleeding

Williams 2018 reported no evidence of a difference in BOP between groups of participants that were exposed to a behavioural intervention with different modes of delivery (MD -1.60%, 95% CI -10.06 to 6.86, favours instructor delivery; 58 participants; Analysis 3.1).

Plaque

Williams 2018 reported no evidence of a difference in plaque between groups of participants that were exposed to a behavioural intervention with different modes of delivery (MD -3.30%, 95% CI -9.31 to 2.71, favours instructor delivery; 58 participants; Analysis 3.2).

Adverse effects

Cleeren 2014 reported that: "No harms befell any participant as a result of participation in the trial".

Behavioural interventions compared with an active control for adults with gingivitis

Given the wide variation in intervention designs, we did not pool data. We have reported a narrative summary of the effect estimates for each outcome. We combined data from the intervention groups for Walsh 1985 when calculating effect estimates for this comparison.

Critical outcomes

For all critical outcomes in this comparison, we were very uncertain of any benefits of behavioural interventions compared to active

control as the certainty of evidence was very low. We downgraded the certainty of evidence for outcomes in this comparison by one level for risk of bias and two levels for imprecision because the evidence was derived from either one very small study or from only two very small studies, and some effect estimates indicated no benefit for either group. See [Summary of findings 2](#).

Gunpinar 2022 [79, 80] and Holloway 2021 reported some data for which we were unable to calculate effect estimates (reported as median values, minimum to maximum); see [Supplementary material 12](#). We did not include these data when assessing the certainty of the evidence.

Bleeding

Daly 2023 reported that participants exposed to a behavioural intervention had less gingival bleeding than those exposed to control (MD -29.00%, 95% CI -36.87 to -21.13; 83 participants). Hartono 2024 similarly reported a benefit in favour of the intervention for participants with gingivitis (MD -23.26%, 95% CI -32.46 to -14.06; 22 participants). Walsh 1985 also reported that participants in both intervention groups had less gingival bleeding than those in the control group (comparison with "toothpick with bleeding feedback" intervention: MD -31.00%, 95% CI -45.19 to -16.81; 24 participants; comparison with "toothpick only" intervention: MD -25.00%, 95% CI -36.65 to -13.35; 24 participants). However, Holloway 2022 reported very imprecise data (MD -42.00%, 95% CI -86.67 to 2.67; 22 participants). See Analysis 4.1.

In addition, Holloway 2021 reported the difference in change in each group, with separate data for participants with low gingivitis (gingival bleeding < 20%) or high gingivitis (gingival bleeding ≥ 20%) and adjustment for clusters. In this study, there was a larger improvement in gingival bleeding for participants with low gingivitis exposed to the behavioural intervention (MD -21.40, 95% CI -42.02 to -0.78; 279 participants). However, this improvement was not evident for participants with high gingivitis (MD -2.80, 95% CI -18.50 to 12.90; 259 participants). See Analysis 4.2.

Gingival inflammation

Only Holloway 2022 reported data for gingival inflammation, measured using a modified gingival index score in which lower scores indicate less inflammation. In this study, there was little or no difference between intervention and control groups (MD -0.70, 95% CI -1.66 to 0.26, favours intervention; 22 participants; Analysis 4.3).

Other clinical markers of periodontal disease

Plaque

This outcome was reported either as a percentage (Analysis 4.4) or using the 0-to-2 Silness and Løe plaque index score [76] in which lower scores indicate less plaque (Analysis 4.5).

When reported as a percentage, Holloway 2022 reported data that were imprecise for levels of plaque (MD -13.00%, 95% CI -28.47 to 2.47; 22 participants). In Walsh 1985, data were similarly imprecise for comparisons with both types of intervention (comparison with "toothpick and bleeding feedback" intervention: MD -9.00%, 95% CI -21.41 to 3.41, favours intervention; 24 participants; comparison with "toothpick only" intervention: MD -3.00%, 95% CI -11.49 to 5.49, favours intervention; 24 participants).

When measured on an index scale, Daly 2023 found little or no difference between intervention and control groups in plaque scores (MD -0.09, 95% CI -0.32 to 0.14, favours intervention; 83 participants).

Depth of gingival pockets

Only one study reported depth of gingival pockets (Hartono 2024), with data indicating a benefit in favour of participants who were exposed to a behavioural intervention (MD -1.33 mm, 95% CI -1.73 to -0.93; 22 participants; Analysis 4.6).

Self-reported measures of oral health-related behaviours

Only one study reported data for this outcome. Holloway 2022 found little or no difference between groups in the number of participants brushing teeth at least twice a day (RR 1.10, 95% CI 0.86 to 1.40, favours intervention; 22 participants; Analysis 4.7).

Adverse effects

No studies reported adverse effects related to the intervention. However, we noted that four studies reported that there were no adverse events during the study period (Daly 2023; Hartono 2024; Holloway 2021; Holloway 2022).

Important outcomes

Self-reported beliefs about and attitudes to oral health-related behaviour

Only one study reported data for this outcome. Gunpinar 2022 measured participant motivation and found that participants were slightly more motivated to perform oral hygiene after exposure to the behavioural intervention (MD 0.12, 95% CI 0.03 to 0.21; 50 participants; Analysis 4.8).

Behavioural interventions compared with an alternative behavioural intervention for adults with gingivitis

Only Walsh 1985 provided data for this comparison. No outcome data were available for gingival inflammation, clinical markers of periodontal disease other than plaque, self-reported measures of oral health-related behaviours, or self-reported beliefs and attitudes to oral health-related behaviour.

Critical outcomes

Bleeding

Walsh 1985 reported little or no difference between intervention groups in gingival bleeding at three months (MD -6.00%, 95% CI -22.52 to 10.52, favours "toothpick with bleeding feedback" intervention; 24 participants; Analysis 5.1).

Other clinical markers of periodontal disease

Plaque

Walsh 1985 reported little or no difference between intervention groups in plaque at three months (MD -6.00%, 95% CI -19.20 to 7.20; 24 participants; Analysis 5.2).

Equity assessment

We did not plan to assess equity-related issues in this review. However, we collected baseline for studies according to the PROGRESS-Plus characteristics described by O'Neill and colleagues [25]. Most studies only reported age and sex of included

participants, and these were reasonably comparable between groups; the average age of participants was older than 30 years.

Of the eight studies reporting race or ethnicity, we noted that six studies included all (or a majority of) participants who were described as white (or Caucasian). In another study, most participants were described as being either white or black, with a smaller number described as being 'Hispanic' or 'other' for race or ethnicity status, and in a study conducted in Sweden, most participants were described as being Swedish. Ten studies reported levels of education, but with varying categories which were not necessarily comparable with one another, and only two studies reported socioeconomic status in terms of income. No studies reported information related to place of residence, occupation, religion, or social capital. A summary of these equity-related characteristics of study participants is provided in [Supplementary material 13](#).

DISCUSSION

Summary of main results

For adults with periodontitis, we found very low-certainty evidence (from 17 RCTs, involving 892 participants) for all outcomes. It is therefore unclear if adults with periodontitis who received behavioural interventions are likely to have fewer clinical markers of disease or report more improved oral hygiene behaviours than those who received an active control (typically, a usual care approach to oral hygiene advice).

Similarly, for adults with gingivitis, we found very low-certainty evidence (from six RCTs, involving 958 participants), and it is unclear if adults receiving a behavioural intervention are more likely to see an improvement than those receiving control.

Limitations of the evidence included in the review

Studies were often either small or very small, and we downgraded the certainty of evidence based on sample sizes, or because the CIs in some effect estimates included the possibility of both benefit or no benefit of the intervention (imprecision). We also downgraded the certainty of the evidence because we assessed at least one domain in all studies as being at either unclear or high risk of bias, and the direction of effect between studies sometimes differed (inconsistency). See [Summary of findings 1](#); [Summary of findings 2](#).

The intervention designs largely related to two behaviours: toothbrushing and interdental cleaning. However, approaches to improving these behaviours differed widely between intervention designs in their dose intensity, frequency, and delivery mode. Due to this heterogeneity, it was not meaningful for us to combine the data quantitatively, and this review is therefore limited to a narrative summary of individual study results. The variety of approaches used by researchers indicates a lack of consensus for intervention design in this context. Whilst we used a recognised taxonomy to describe the interventions according to their behaviour change techniques (BCTs) [7], coding was sometimes limited by lack of information reported. In addition, our review was limited in its capacity to unpick the differences in approaches and explore whether there are differences in effectiveness between approaches. Reporting of the interventions could be improved upon; specifically, details of the design, delivery, and fidelity of interventions were often poor or absent.

We note that some studies included compliance (defined by regular dental attendance) as an eligibility criterion. This must be considered when applying findings to general populations who may, on the whole, exhibit both lower levels of compliance with dental attendance and oral hygiene behaviours. A large portion of the studies were conducted with small groups of participants, limiting the precision of their findings.

We found fewer studies involving people in the supportive phase of periodontal treatment than in the active phase. Again, this limits the generalisability of findings from individual studies for this population group, in which different approaches may be required to establish long-term behaviour change. We also found few studies with follow-up times beyond six months. Only two studies included a follow-up later than six months, both reporting data at 12 months. Due to the importance of long-term maintenance of oral hygiene behaviours, studies with longer follow-up are important to demonstrate a meaningful effect.

Limitations of the review processes

We conducted the review according to Cochrane methodological standards. We made some deviations to the protocol as described in [Supplementary material 8](#), which were intended to improve the robustness and usability of the review findings. Of note, we limited the criteria to only studies involving a clearly defined population with periodontal disease. Whilst we are aware of the availability of published studies involving mixed populations (with and without disease), it was not our intention to draw conclusions about interventions targeting a more general population. We also limited the review to interventions that included BCTs rather than any type of intervention related to oral hygiene advice; this change allowed us to more specifically address the review objectives.

Agreements and disagreements with other studies or reviews

This review is an update to a review published by Renz and colleagues in 2007 [81]. We have adapted the eligibility criteria, meaning that there are differences in the studies included. The review by Renz and colleagues included four studies and concluded there was limited evidence for these types of interventions, but that there was some promise and further research was warranted. The current review includes 25 studies, demonstrating that research in this area has increased. However, the quality of this research is limited, meaning conclusions remain uncertain.

Other reviews in this area have not been able to conclude that there is strong evidence in support of these types of interventions [82, 83]. Two reviews cautiously reported that interventions using elements of social cognitive theory, including self-efficacy [84] and goal-setting, planning, self-monitoring, and feedback in particular, may have promise [85]. Some reviews have focused on specific types of interventions. Of note, there are several systematic reviews of motivational interviewing for supporting oral hygiene [83, 86, 87, 88, 89]. Those reviews are in agreement that there is currently insufficient evidence for motivational interviewing in this context; this is congruent with the conclusions of our review.

AUTHORS' CONCLUSIONS

Implications for practice

Based on very low-certainty evidence, we are unable to conclude whether behavioural interventions are better than usual care advice at improving the oral hygiene of adults with periodontitis or gingivitis. Given the wide variation in intervention designs, we did not pool any data in this review.

Implications for research

Significantly more studies have been published since the last related version of this Cochrane review, indicating the important interest in behavioural techniques to improve oral hygiene in this population. However, confidence in the certainty of the existing evidence remains very low.

We recommend that future studies continue to focus on toothbrushing and interdental cleaning. Interventions should be developed based on techniques that have a clear mechanism of action for change; using the behaviour change techniques taxonomy v1 is an appropriate starting point [7]. Studies that are sufficiently large and ideally include multiple centres, such as Holloway 2021, are more likely to provide generalisable data. We advocate for studies to include robust methodological and reporting standards with trial reports adhering to CONSORT guidelines [90], with intervention content transparently reported, which would improve understanding of the intervention and replication in other studies or implementation as appropriate. We encourage the use of reporting guidelines such as Workgroup for Intervention Development and Evaluation Research (WIDER) [91] or COMPASS [20]. Future trials should use the core outcome set developed for periodontal trials [92]; additionally, this field would benefit from consensus on the best measures for the collection of self-reported oral hygiene behaviours.

A number of the control groups in studies in this review experienced 'usual care' which could be coded using the BCT taxonomy, meaning that usual care can be described as a behavioural intervention. It may be fruitful to formalise 'usual care' as behavioural interventions to standardise patient care and optimise the role of the dental team in supporting patient behaviour change.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD012049.pub2](https://doi.org/10.1002/14651858.CD012049.pub2).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Analyses

Supplementary material 7 Data package

Supplementary material 8 Differences between protocol and review

Supplementary material 9 Template data extraction form

Supplementary material 10 BCT taxonomy (v1): 93 hierarchically-clustered techniques

Supplementary material 11 Intervention characteristics and behaviour change techniques used in each study

Supplementary material 12 Data not included in review analyses

Supplementary material 13 Equity-related information according to PROGRESS-Plus in the included studies

ADDITIONAL INFORMATION

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- Sign-off Editor (final editorial decision): Fang Hua, School and Hospital of Stomatology, Wuhan University;
- Managing Editor (selected peer reviewers, collated peer-reviewer comments, provided editorial guidance to authors, and edited the article): Marwah Anas El-Wegoud, Cochrane Central Editorial Service;
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Contributions of authors

Co-ordination of review: SL, LO'M

Content expertise: PA, MB, ME, GK, PP, HR, P-MJ-S

Identification of studies: PA, MB, SL, LO'M, PP, PR, P-MJ-S

Data extraction: SL, LO'M, PR, P-MJ-S

Risk of bias assessment: SL, LO'M, PR

Data input and analysis: SL, LO'M, PR

Writing of discussion and conclusions: PA, MB, ME, GK, SL, LO'M, PP, HR, PR, P-MJ-S

Declarations of interest

LO'M is a former editor for the Cochrane Oral Health group; she was not involved in the editorial process for this review. No commercial or non-commercial interests relevant to this review.

SL is the former Deputy Co-ordinating Editor of the Cochrane Bone, Joint and Muscle Trauma Group; she was not involved in the

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PP is Editor for the *British Dental Journal*. He received payment to deliver a lecture by GlaxoKlineSmith in March 2024.

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ME has no commercial or non-commercial interests relevant to this review.

HR has no commercial or non-commercial interests relevant to this review.

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Registration and protocol

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Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library (see [Supplementary material 6](#) and [Supplementary material 7](#): full search strategies for each database; full citations of each unique report for all studies (included, ongoing or awaiting classification, or key excluded at the full text screen) in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; and analysis data, including individual data rows. Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods.

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ADDITIONAL TABLES

Table 1. Overview of included studies: behavioural interventions compared with active control for adults with periodontitis

Studies including adults in the active phase of treatment					
Study	Sample size ^a	Intervention (BCT cluster type)	Control (BCT cluster type)	Outcomes ^b	Final follow-up
Alasmari 2022	39	Shaping knowledge, comparison of the behaviours, associations, and repetition and substitution	Shaping knowledge	GI Plaque	1 month
de Souza 2002	60	Goals and planning, feedback and monitoring, shaping knowledge, natural consequences, comparison of behaviour, associations, repetition and substitution, comparison of outcomes, and reward and threat	Shaping knowledge, comparison of behaviour, and repetition and substitution	Bleeding ^c Plaque ^c	3 months
Fiebranz 1989	50	Feedback and monitoring, shaping knowledge, comparison of behaviour, and repetition and substitution	Shaping knowledge, comparison of behaviour, and repetition and substitution	Bleeding ^d Plaque ^d	6 months
Godard 2011	51	Goals and planning, feedback and monitoring, shaping knowledge, natural consequences, comparison of behaviour, repetition and substitution, antecedents, and self-belief	Shaping knowledge, comparison of behaviour, repetition and substitution, and antecedents	Plaque	1 month
Hartono 2024	18	Feedback and monitoring, shaping knowledge, comparison of behaviour, associations, and repetition and substitution	Shaping knowledge, comparison of behaviour, and repetition and substitution	Bleeding PPD	3 months
Järvinen 2020	28	Goals and planning, shaping knowledge, comparison of behaviour, and repetition and substitution	Shaping knowledge, comparison of behaviour, and repetition and substitution	Bleeding PPD ^c CAL ^c Self-reported measures of oral health-related behaviours ^c	6 months
Jönsson 2009	113	Goals and planning, feedback and monitoring, shaping knowledge, comparison of behaviour, and repetition and substitution	Shaping knowledge, comparison of behaviour, and repetition and substitution	Bleeding GI Plaque PPD Self-reported measures of oral health-related behaviours	12 months
Kakudate 2009	38	Goals and planning, feedback and monitoring, shaping knowledge, comparison of the behaviour, repetition and substitution	Shaping knowledge, comparison of behaviour, and repetition and substitution	Plaque Self-reported measures of	3 weeks

Table 1. Overview of included studies: behavioural interventions compared with active control for adults with periodontitis (Continued)

		tion and substitution, and reward and threat	ior, and repetition and substitution	oral health-related behaviours	
Lin 2019	330	Goals and planning, shaping knowledge, natural consequences, and antecedents	Shaping knowledge, natural consequences, and antecedents	Self-reported measures of oral health-related behaviours ^e	6 weeks
Little 1997	107	Goals and planning, feedback and monitoring, social support, shaping knowledge, comparison of the behaviour, associations, repetition and substitution, reward and threat, antecedents and scheduled consequences	Shaping knowledge, comparison of behaviour, and repetition and substitution	Bleeding ^d GI ^d Plaque ^d PPD ^d CAL ^d	4 months
Philippot 2005	33	Goals and planning, feedback and monitoring, shaping knowledge, natural consequences, comparison of behaviour, repetition and substitution, and antecedents	Feedback and monitoring, shaping knowledge, comparison of behaviour, repetition and substitution, and antecedents	Plaque	1 month
Stenman 2012	44	Goals and planning, shaping knowledge, comparison of behaviour, associations, repetition and substitution, antecedents, identity, and self-belief	Shaping knowledge, comparison of behaviour, and repetition and substitution	Bleeding Plaque	6 months
Zhang 2018	105	Shaping knowledge, natural consequences, comparison of behaviour, repetition and substitution	Shaping knowledge, comparison of behaviour, and repetition and substitution	None	n/a
Studies including adults in the supportive phase of treatment					
Study	Sample size ^a	Intervention (BCT cluster type)	Control (BCT cluster type)	Outcomes ^b	Final follow-up
Arnett 2023	60	Shaping knowledge, comparison of the behaviours, repetition and substitution, reward and threat, and scheduled consequences	Shaping knowledge, comparison of behaviour, and repetition and substitution	Bleeding GI Plaque	12 months
Baab 1986	33	Feedback and monitoring, shaping knowledge, comparison of behaviour, repetition and substitution, antecedents	Shaping knowledge, comparison of behaviour, and repetition and substitution	Bleeding ^c Plaque ^c	6 months
Brand 2013	56	Goals and planning, shaping knowledge, comparison of behaviour, repetition and substitution, comparison and outcomes, reward and threat, identity, and self-belief	Shaping knowledge, comparison of behaviour, and repetition and substitution	Bleeding Plaque PPD	3 months

Table 1. Overview of included studies: behavioural interventions compared with active control for adults with periodontitis (Continued)

Jönsson 2006	37	Goals and planning, feedback and monitoring, shaping knowledge, natural consequences, comparison of behaviour, and repetition and substitution	Shaping knowledge, comparison of behaviour, and repetition and substitution	Bleeding GI Plaque PPD Self-reported measures of oral health-related behaviours	3 months
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Abbreviations: **BCT:** behaviour change techniques, using cluster labels according to Michie and colleagues [7]; **CAL:** clinical attachment loss; **GI:** gingival inflammation; **n/a:** not applicable; **PPD:** probing pocket depth

^aNumber randomised to the intervention and control group.

^bOutcomes reported by study authors considered to be critical in this review.

^cUnable to extract data.

^dData reported without distribution values and not used to calculate effect estimates in this review.

^eData not included in the review owing to high risk of bias in several risk of bias domains.

Table 2. Overview of included studies: behavioural interventions compared to active control for adults with gingivitis

Study	Sample size ^a	Intervention (BCT cluster type)	Control (BCT cluster type)	Outcomes ^b	Final follow-up
Daly 2023	85	Feedback and monitoring, shaping knowledge, comparison of behaviour, associations, repetition and substitution, and antecedents	Shaping knowledge, comparison of behaviour, and repetition and substitution	Bleeding Plaque	6 months
Gunpinar 2022	60	Feedback and monitoring, shaping knowledge, natural consequences, comparison of behaviour, and repetition and substitution	Shaping knowledge, comparison of behaviour, and repetition and substitution	Bleeding ^c Plaque ^c	6 months
Hartono 2024	22	Feedback and monitoring, shaping knowledge, comparison of behaviour, associations, and repetition and substitution	Shaping knowledge, comparison of behaviour, and repetition and substitution	Bleeding Depth of gingival pockets	3 months
Holloway 2021	733	Goals and planning shaping knowledge, natural consequences, comparison of behaviour, repetition and substitution, comparison of outcomes, and antecedents	Shaping knowledge, comparison of behaviour, and repetition and substitution	Bleeding Plaque ^c Self-reported measures of oral health-related behaviours ^c	3 months
Holloway 2022	22	Feedback and monitoring, shaping knowledge, natural consequences, comparison of behaviour, repetition and substitution, antecedents	Shaping knowledge, comparison of behaviour, and repetition and substitution	Bleeding GI Plaque Self-reported measures of	1 month

Table 2. Overview of included studies: behavioural interventions compared to active control for adults with gingivitis (Continued)

				oral health-re- lated behav- iours	
Walsh 1985	36	Feedback and monitoring, shaping knowledge, natural consequences, comparison of behaviour, repetition and substitution, antecedents ^d	Feedback and monitoring, shaping knowledge, comparison of behaviour, repetition and substitution, antecedents	Bleeding Plaque	3 months

Abbreviations: **BCT:** behaviour change techniques, using cluster labels according to Michie and colleagues [7]; **GI:** gingival inflammation

^aNumber randomised to the intervention and control group.

^bOutcomes reported by study authors considered to be critical in this review.

^cData reported as median and interquartile ranges (or minimum to maximum), and not used to calculate effect estimates in this review.

^dStudy included two intervention groups. Whilst both groups were shown how to use a toothpick, participants in one intervention were taught that bleeding could be used to provide feedback on their performance.

Table 3. Outcomes reported using categorical data

Comparison: behavioural intervention vs active control in adults with periodontitis		
Outcome: self-reported measures of oral health-related behaviour	Results from study report	Study
Performance of oral hygiene	Intervention group – much better/better, n: 45/5 Control group – much better/better, n: 40/15	Jönsson 2009
Compliance to skills obtained during treatment (daily)	Intervention group – daily/several times a week/some times a week, n: 44/4/2 Control group – daily/several times a week/some times a week, n: 37/17/1	Jönsson 2009
Outcome: self-reported beliefs about and attitudes to oral health-related behaviour	Results from study report	Study
Do you think that you yourself can contribute to the health of teeth and gums?	Intervention group – very much/a lot/quite a bit/little/nothing, n: 14/4/2/0/0 Control group – very much/a lot/quite a bit/little/nothing, n: 7/7/12/4/0	Fiebranz 1989
Do you have the impression that the pretreatment (the intervention) has helped you?	Intervention group – helped very well/helped well/helped some/hardly helped/didn't help at all, n: 14/6/0/0/0 Control group – helped very well/helped well/helped some/hardly helped/didn't help at all, n: 6/8/14/2/0	Fiebranz 1989
How tedious/exhausting/annoying did you find the treatment?	Intervention group – extremely troublesome/very laborious and exhausting/hard and exhausting/hardly laborious and exhausting/not laborious and exhausting, n: 0/0/3/9/8	Fiebranz 1989

Table 3. Outcomes reported using categorical data *(Continued)*

	Control group – extremely troublesome/very laborious and exhausting/hard and exhausting/hardly laborious and exhausting/not laborious and exhausting, n: 0/1/9/10/10	
Do you think you can maintain the condition of your teeth without further guidance and care?	Intervention group – yes/no/unsure, n: 2/13/4	Fiebranz 1989
	Control group – yes/no/unsure, n: 4/7/19	
Does the time required for the goal of improvement appear to be justified by your oral hygiene?	Intervention group – yes/no/too much expense, n: 20/0/0	Fiebranz 1989
	Control group – yes/no/too much expense, n: 24/3/3	
Satisfaction with oral health	Intervention group – much more satisfied/more satisfied/no difference, n: 35/15/0	Jönsson 2009
	Control group – much more satisfied/more satisfied/no difference, n: 31/23/1	
Likely to maintain habits	Intervention group – very likely/likely, n: 46/5	Jönsson 2009
	Control group – very likely/likely, n: 39/16	

n: number of participants